Deeba Afridi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
Carmel Convent Senior Secondary School	Class XII (CBSE)	2018	74.00%
Carmel Convent Senior Secondary School	Class X (CBSE)	2016	9.60%

TECHNICAL SKILLS

Programming Languages: C, C++, Python, SQL

Compilers & Systems: Clang, LLVM, MLIR, PyTorch Compilation, ML Compilers, Static Analysis, IR Transformations, AST/IR Pass Development, Code Optimization, Code Compliance

GPU & Parallel Computing: CUDA, CUDA Kernels, CUTLASS, GPU Performance Optimization

AI/ML: NumPy, Pandas, PyTorch, Machine Learning, Deep Learning

Tools & Platforms: Git, GitHub, Linux, Nsight Systems, Nsight Compute, PyTorch Profiler, MySQL

WORK EXPERIENCE

Research Assistant | *IIT Hyderabad*

- Designed and evaluated **Clang-based LLVM static analysis tooling** for detecting C/C++ code compliance violations in safety-critical systems for an industry partner, with rule-specific analysis and compliance test-suite validation.
- Developed and integrated a **graph transformation** into the PyTorch compilation pipeline to enable larger computational graphs and broader CUDA Graph capture, achieving **1.5×-2× speedup** across the evaluated models.
- Implemented **compute- and memory-intensive CUDA kernels**, applying warp specialization to overlap memory operations with computation, improving GPU resource utilization and overall kernel performance.

Assistant System Engineer | *TCS, Indore*

- Developed and integrated **J2EE/Servlet backend APIs**, maintained **database schemas**, and implemented **data validation** to support business requirements while ensuring data consistency, correctness, and reliable application processing.
- Built **JUnit test cases** achieving **90% code coverage**, validating application functionality and improving overall software reliability.

PROJECTS

PyTorch FX Graph Optimization Framework | *Python, PyTorch, TorchDynamo, PyTorch Profiler*

- Built a **CFG-based bytecode transformation framework** and integrated it into **TorchDynamo**, using instruction rewriting across frame conversion, bytecode transformation, and inlining/tracing to eliminate torch.compile graph breaks before tracing.
- Evaluated the framework on **Hugging Face models**, eliminating graph breaks to enable larger FX graphs and broader CUDA Graph capture, improving compilation continuity and achieving **1.5×-2× speedup** across evaluated models.

MLIR-CUTLASS Convolution Offloading Pipeline | *C++, MLIR, Torch-MLIR, CUDA, CUTLASS, PyTorch, Python*

- Built an end-to-end **MLIR-CUTLASS compiler pipeline**, evaluated on **ResNet50**, lowering all **53 convolution** operations to hand-tuned **CUTLASS kernels** while compiling the remaining operators to MLIR-generated GPU kernels.
- Developed an MLIR analysis/transformation framework to trace **linalg dependencies** and lower convolutions to CUTLASS calls, using a **CUDA bridge** for memrefs, CUTLASS buffers, and **NCHW-NHWC conversion**, generating 53 CUTLASS calls and 422 GPU launches.

Multi-Object Scene Classification using CNNs | *Python, PyTorch, NumPy*

- Designed a **dual-head residual CNN** for **18K** four-object scenes with separate cube/sphere heads and a **composite BCE loss**.
- Applied **Sobel edge gradients** to suppress color/lighting variations and emphasize shape-focused feature learning.
- Used **CoordConv** to inject explicit spatial information, improving spatial awareness under occlusion and scale variation.
- Benchmarked **ViT, Slot Attention, and color masking** against the proposed dual-head CNN, which achieved **75.88%** validation accuracy with **4-pass TTA** and up to **10.88** percentage-point improvement.

Blocking and Non-Blocking Concurrent Synchronization | *C++, Atomics, Mutexes*

- Implemented and benchmarked **blocking synchronization**, measuring average/worst-case lock-acquisition time and throughput across varying thread counts and critical-section entry frequencies to analyze contention effects.
- Implemented **non-blocking atomic snapshots** over **MRMW registers** using stamped values and repeated collect/compare checks to detect concurrent modifications and ensure consistent multi-reader/multi-writer snapshots.
- Evaluated scalability across **4-64 threads** and writer/snapshot ratios from **1-10**, comparing average- and worst-case update, scan, and combined execution timings under increasing concurrency.

Parallel BFS on GPU with Overlapped Computation and Data Transfer | *C++, CUDA, CUDA Streams, CUDA Graphs, Python*

- Implemented three progressively optimized **CUDA parallel BFS variants** for k-NN graphs exceeding GPU memory capacity.
- Overlapped runtime **host-to-device transfers** with kernel execution using **4 concurrent CUDA Streams**, enabling transfer-compute concurrency and reducing per-level launch overhead through **CUDA Graphs**.
- Benchmarked the progressively optimized implementations across seven k-NN graph sizes up to **500K nodes**, achieving up to **1.94× speedup** over the baseline and reducing runtime by **~40%** on the largest graphs.

GPU GEMM Optimization: Pipeline Staging and Warp Specialization | *C++, CUDA, Nsight Compute*

- Designed **software-pipelined** CUDA GEMM kernels with producer-consumer warp specialization to overlap shared-memory prefetch and computation, achieving **1.48×-1.63× speedup** over tiled kernels across 2MM, 3MM, and GEMM benchmarks.
- Developed a **warp-specialized FP32/FP64 kernel** with warp-based scheduling across compute pipelines, achieving up to **1.52× speedup** on synthetic benchmarks and profiling pipeline utilization and warp behavior using **Nsight Compute**.
- Implemented a **hardware-pipelined** CUDA GEMM kernel using asynchronous global-to-shared transfers to overlap data movement with computation, achieving up to **~18% performance improvement** in pipeline benchmarks.

RELEVANT COURSES

Program Analysis and Compiler Optimization | Topics in Compiler Optimization | GPU Programming | Parallel & Concurrent Programming | Advanced Data Structures & Algorithms | Machine Learning | Deep Learning | Compilers | Operating Systems | Computer Organization and Architecture | Database Management Systems | Computer Networks

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

5th among 109 teams - ML Hackathon | Qualified **Logical Thinking & Smart Programming** Examination in reasoning & programming

2nd Prize among 100+ teams - Inter-College PPT Competition | Completed **Rotary Youth Leadership Awards** - Leadership & teamwork

Teaching Assistant - Program Analysis & Compiler Optimization, Compilers I, GPU Programming; conducted course/lab sessions and prepared & graded assignments/quizzes | **Core Organizing Team Member** - Annual Function planning & coordination

Rotary Club Representative - College events & community initiatives | **Cabinet Member** - Led student house, events & competitions